

Algebra II Topic Cheatsheet

Comprehensive Review

Polynomials and Factoring

$$a^2 - b^2 = (a - b)(a + b)$$

$$a^2 \pm 2ab + b^2 = (a \pm b)^2$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2), \quad a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

- Factor out the GCF first.
- Use grouping for four terms.
- For $ax^4 + bx^2 + c$, substitute $u = x^2$.
- For quartics, look for symmetric factors of the form $(x^2 + px + q)(x^2 - px + r)$.

Rational Root Theorem and Division

If

$$f(x) = a_n x^n + \cdots + a_1 x + a_0$$

has a rational root $\frac{p}{q}$ in lowest terms, then

$$p \mid a_0, \quad q \mid a_n$$

Possible rational roots:

$$\pm \frac{\text{factors of constant term}}{\text{factors of leading coefficient}}$$

Remainder Theorem:

$$f(r) = \text{remainder on division by } x - r$$

Factor Theorem:

$$f(r) = 0 \iff x - r \text{ is a factor}$$

Synthetic division by $x - r$ uses the number r .

Zeros and Graph Behavior

- Real zeros give x -intercepts.
- Even multiplicity: touches and turns.
- Odd multiplicity: crosses.
- Complex roots of polynomials with real coefficients come in conjugate pairs.
- End behavior comes from degree and leading coefficient.

even degree, positive LC	both ends up
even degree, negative LC	both ends down
odd degree, positive LC	left down, right up
odd degree, negative LC	left up, right down

Rational Functions

For

$$f(x) = \frac{p(x)}{q(x)}$$

- Domain excludes zeros of $q(x)$.
- Vertical asymptotes occur at non-canceled zeros of $q(x)$.
- Holes occur when a factor cancels.
- Horizontal asymptotes:

$$\deg p < \deg q \Rightarrow y = 0$$

$$\deg p = \deg q \Rightarrow y = \frac{\text{leading coeff of } p}{\text{leading coeff of } q}$$

$$\deg p > \deg q \Rightarrow \text{use long division}$$

- Slant asymptote occurs when $\deg p = \deg q + 1$.

Intercepts:

x -intercepts from numerator zeros not canceled, y -intercept $f(0)$

Logarithms and Exponentials

$$y = b^x \iff x = \log_b y$$

$$\log_b b^x = x, \quad b^{\log_b x} = x$$

with

$$b > 0, \quad b \neq 1, \quad x > 0$$

$$\log_b(MN) = \log_b M + \log_b N$$

$$\log_b\left(\frac{M}{N}\right) = \log_b M - \log_b N$$

$$\log_b(M^p) = p \log_b M$$

$$\log_b a = \frac{\log a}{\log b} = \frac{\ln a}{\ln b}$$

Solving templates:

$$b^{f(x)} = b^{g(x)} \Rightarrow f(x) = g(x)$$

$$\log_b(f(x)) = \log_b(g(x)) \Rightarrow f(x) = g(x)$$

Check domain: log arguments must be positive.

Growth and decay:

$$y = ab^t, \quad y = ae^{kt}$$

$$k > 0 \Rightarrow \text{growth}, \quad k < 0 \Rightarrow \text{decay}$$

$$\text{doubling time} = \frac{\ln 2}{k}, \quad \text{half-life} = \frac{\ln(1/2)}{k}$$

Quadratics and Conics

Vertex form:

$$y = a(x - h)^2 + k$$

Axis of symmetry:

$$x = h$$

Vertex:

$$(h, k)$$

Maximum or minimum depends on the sign of a .

Conic identification:

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

- circle: $A = C$, $B = 0$
- parabola: one squared term
- ellipse: same sign, unequal coefficients
- hyperbola: squared terms with opposite signs

Standard forms:

$$\begin{aligned}(x - h)^2 + (y - k)^2 &= r^2 \\(x - h)^2 &= 4p(y - k), \quad (y - k)^2 = 4p(x - h) \\ \frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} &= 1 \\ \frac{(x - h)^2}{a^2} - \frac{(y - k)^2}{b^2} &= 1, \quad \frac{(y - k)^2}{a^2} - \frac{(x - h)^2}{b^2} = 1\end{aligned}$$

For hyperbolas:

$$c^2 = a^2 + b^2, \quad e = \frac{c}{a}, \quad \text{asymptotes } y - k = \pm \frac{b}{a}(x - h)$$

or

$$y - k = \pm \frac{a}{b}(x - h)$$

Trigonometry

$$\begin{aligned}180^\circ &= \pi, \quad 360^\circ = 2\pi \\ \sin \theta &= \frac{\text{opp}}{\text{hyp}}, \quad \cos \theta = \frac{\text{adj}}{\text{hyp}}, \quad \tan \theta = \frac{\text{opp}}{\text{adj}} \\ \csc \theta &= \frac{1}{\sin \theta}, \quad \sec \theta = \frac{1}{\cos \theta}, \quad \cot \theta = \frac{1}{\tan \theta}\end{aligned}$$

Core identities:

$$\begin{aligned}\sin^2 \theta + \cos^2 \theta &= 1 \\ 1 + \tan^2 \theta &= \sec^2 \theta, \quad 1 + \cot^2 \theta = \csc^2 \theta\end{aligned}$$

Special values:

$$\sin\left(\frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}\right) = \left(\frac{1}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}\right)$$

$$\cos\left(\frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}\right) = \left(\frac{\sqrt{3}}{2}, \frac{\sqrt{2}}{2}, \frac{1}{2}\right)$$

$$\tan\left(\frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}\right) = \left(\frac{\sqrt{3}}{3}, 1, \sqrt{3}\right)$$

Law of Sines:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Law of Cosines:

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Triangle area:

$$K = \frac{1}{2}ab \sin C$$

Sequences and Series

Arithmetic sequence:

$$a_n = a_1 + (n-1)d$$

$$S_n = \frac{n(a_1 + a_n)}{2}$$

Geometric sequence:

$$a_n = a_1 r^{n-1}$$

$$S_n = \frac{a_1(1 - r^n)}{1 - r} \quad (r \neq 1)$$

Infinite geometric series:

$$S_\infty = \frac{a_1}{1 - r} \quad (|r| < 1)$$

Sigma notation:

$$\sum_{k=1}^n a_k$$

$$\text{factorials: } n! = n(n-1) \cdots 1$$

$${}_nP_r = \frac{n!}{(n-r)!}, \quad {}_nC_r = \frac{n!}{r!(n-r)!}$$

Complex Numbers and Polar

$$i^2 = -1, \quad \sqrt{-a} = i\sqrt{a}$$

If

$$z = a + bi,$$

then its conjugate is

$$a - bi$$

and

$$(a + bi)(a - bi) = a^2 + b^2$$

Polar form:

$$z = r(\cos \theta + i \sin \theta) = r \operatorname{cis} \theta$$

Polar to Cartesian:

$$x = r \cos \theta, \quad y = r \sin \theta$$

Cartesian to polar:

$$r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x}$$

De Moivre:

$$[r(\cos \theta + i \sin \theta)]^n = r^n(\cos n\theta + i \sin n\theta)$$

Nth roots of $r(\cos \theta + i \sin \theta)$:

$$z_k = r^{1/n} \left(\cos \frac{\theta + 2\pi k}{n} + i \sin \frac{\theta + 2\pi k}{n} \right), \quad k = 0, 1, \dots, n-1$$

Quick Algebra II Checks

- When solving, check for extraneous roots from logs and rational equations.
- Use exact values whenever possible.
- State rounding only when asked.
- For rational equations, multiply by the LCD only after listing domain restrictions.